An Expert-Following Strategy for Polymarket

Methodology and Backtest Results

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Executive Summary

- We test a settlement-only Polymarket strategy that follows historically skilled traders.
- Trader expertise is estimated from past settled markets using semantic similarity between market questions.
- The current strategy trades only the high-probability "stable" bucket: entry price > 0.8.
- The latest full-market-scale backtest covers roughly 3.11 years of trade history.
- Final portfolio value is evaluated using total equity, not only cash balance.

Data Sources and Coverage

- Market metadata is fetched from the Polymarket Gamma API.
- Public historical trades are fetched from the Polymarket Data API.
- Cached market metadata universe:
  - 299,318 market records.
- Cached trade sample used for the reported results:
  - `trade_market_count = 3000`: top-volume markets only.
  - `per_market_trades_limit = 400`: at most 400 sampled trades per market.
  - 1,195,104 normalized trade rows.
  - 2,993 trade-covered markets.
  - 2,516 trade-covered markets with available resolution labels.
- The reported backtests do not use complete trade history for all 299,318 markets.
- Local cache is used for markets, trades, and embeddings to make repeated runs practical.

Important Modeling Assumption

Settlement-only PnL

We do not model mid-market exits, stop-losses, or selling before resolution. Every position is evaluated by final settlement or by mark-to-last-known trade price if still open.

- This differs from stock-style backtests.
- It matches the intended research question: buy a prediction market outcome and hold until final resolution.
- Reported `total_equity` is preferred over `final_balance`, because open positions may lock cash.

Anti-Leakage Design

- The event stream is processed chronologically.
- At time t, trader skill uses only markets settled no later than t.
- Current market resolution is never used to generate current signals.
- A signal is executed only after a configurable delay.
- Time-to-resolution filters are computed using market metadata available at entry time.

Trader Market Score

For trader u in settled market m:

$$\text{score}_{u,m} = \text{clip}\left(\frac{\text{PnL}_{u,m}}{\text{notional}_{u,m}}, -1, 1\right)$$

where:

- `notional` is absolute traded dollar volume.
- `PnL` is computed from cash flows plus final settlement payout.
- Scores are clipped to reduce extreme unit-return outliers.

Semantic Similarity Between Markets

Each market question is embedded into a vector:

$$e_m = \text{Embed}(\text{question}_m)$$

Similarity is cosine similarity:

$$s(m, n) = \max\left(0, \cos(e_m, e_n)\right)$$

Current implementation supports:

- `sentence_transformers`: currently `BAAI/bge-large-en-v1.5`.
- Larger `Qwen3-Embedding` experiments did not materially improve backtest results.
- `hashing`: faster fallback for parameter sweeps.

Trader Skill on a New Market

For trader u and target market q at time t:

$$\widehat{\text{skill}}_{u,q,t} = \frac{\sum_{m \in \mathcal{H}_{u,t}} s(q,m) \cdot \text{notional}_{u,m} \cdot \text{score}_{u,m}}{\sum_{m \in \mathcal{H}_{u,t}} s(q,m) \cdot \text{notional}_{u,m}}$$

where $\mathcal{H}_{u,t}$ contains only markets already settled by time t.

A trader is considered skilled in the current flow if:

$$\widehat{\text{skill}}_{u,q,t} \geq \texttt{skill\_threshold}$$

Direction Consensus Signal

Each skilled trade contributes direction-weighted flow:

$$w_i = \widehat{\text{skill}}_i \cdot \text{volume}_i$$

Direction ratios:

$$R_{\text{YES}} = \frac{\sum_{i \in \text{YES}} w_i}{\sum_i w_i}, \quad R_{\text{NO}} = \frac{\sum_{i \in \text{NO}} w_i}{\sum_i w_i}$$

A signal is triggered when:

$$\max(R_{\text{YES}}, R_{\text{NO}}) \geq \texttt{consensus\_threshold}$$

Current Strategy Rules

Baseline fixed-fraction configuration:

- Trade only the stable bucket: entry token price > 0.8.
- Lottery bucket is disabled.
- Direction must pass expert consensus threshold.
- Entry must satisfy:

$$5 < \text{days to resolution} < 40$$

- Stable bucket position size:

$$\text{notional} = 0.2 \times \text{current cash balance}$$

- Maximum two entries per market.

Target Exposure Sizing

Alternative sizing experiment:

$$\text{target deployed} = 0.97 \times \text{total equity}$$

$$\text{available cash} = \text{cash} - 0.05 \times \text{total equity}$$

$$\text{annualized edge} = \max(\text{confidence} - \text{entry price}, 0) \times \frac{365}{\text{days to resolution}}$$

$$\text{order notional} = \text{remaining target budget} \times \text{clip}(0.25 \times \text{annualized edge}, 0.05, 0.35)$$

Backtest Framework

Market metadata + public historical trades

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Question embeddings and market similarity

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Historical settlement PnL → trader skill

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Skilled-trader directional consensus

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Delayed execution + risk and sizing rules

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Settlement-only PnL + total equity curve

Experiment Setup and Headline Results

Metric	Fixed 20%	Target Exposure
Trade period	2023-03-16 to 2026-04-24	
Duration	1134.95 days / 3.107 years	
Market metadata records	299,318	
Trade-covered markets	2,993	
Trade sampling config	top 3,000 markets, max 400 trades each	
Historical trades sampled	1,195,104	
Unique traders	330,576	
Strategy trades	513	514
Win rate	99.03%	99.03%
Final total equity	22,600.15	33,220.86
Approx. CAGR	30.00%	47.12%

- These results are not based on complete trade history for all 299,318 markets.

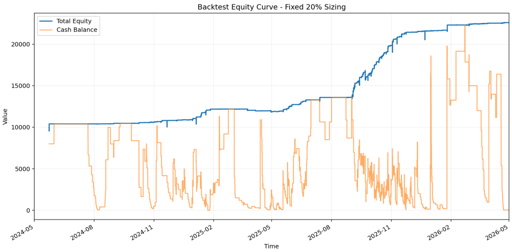
Detailed Results Comparison

Metric	Fixed 20%	Target Exposure
trades	513	514
win_rate	0.990253	0.990272
total_pnl	12,541.4474	22,783.3788
avg_pnl_per_trade	24.4473	44.3256
pnl_volatility	81.3893	195.9001
final_balance	34.2648	1,589.7532
open_positions	27	34
open_notional	22,507.1826	31,193.6256
open_market_value	22,565.8838	31,631.1050
open_unrealized_pnl	58.7013	437.4794
total_equity	22,600.1487	33,220.8582

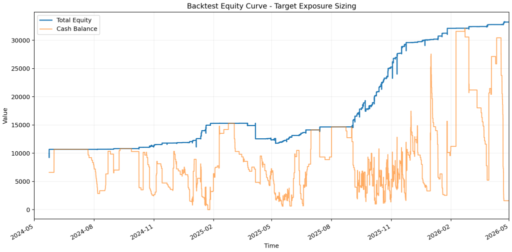
- Target exposure keeps more cash buffer while deploying more total notional.
- The higher ending equity comes with higher realized PnL volatility.

Equity Curve Comparison

Fixed 20% sizing



Target exposure sizing



Result Interpretation

- High win rate comes from trading high-probability stable outcomes.
- Positive total equity indicates that the expert-following filter adds value over this sample.
- Target exposure sizing improves capital deployment and total equity in this run.
- The improvement comes with higher PnL volatility and larger open notional.
- A 1,000-trades-per-market rerun did not improve over the reported 400-trade sample result.
- The important portfolio value metric is total equity, not cash alone.

What Makes This Method Different

Training-free expert following

The strategy does not fit a supervised prediction model or rely on a fixed train/test split. At each event time, it learns from traders who were historically strong on already-settled, semantically similar markets. Market embeddings are fixed representations and are not fitted to backtest outcomes, which reduces classic label-overfitting risk.

Main limitations

The approach still depends on manually selected thresholds, a hand-designed trader-skill score, and a hand-designed sizing rule. It may also weaken if strong traders rotate accounts or if the sampled public trade data misses important market flow.

Implemented but Disabled

The codebase includes several research features that are currently turned off in the reported configuration:

- Lottery bucket: trades with $0.1 \leq p \leq 0.5$, controlled by a small fixed lot size.
- Lottery exposure cap: maximum lottery exposure as a fraction of total capital.
- Dynamic entry price cap: stricter or looser maximum entry price based on consensus confidence.
- Minimum edge filter: requires consensus-implied edge over entry price.
- Alternative trader-skill weighting: configurable historical notional exponent.
- Hashing-vectorizer embeddings: fast fallback for parameter sweeps.

Thank You